Fusing Hyperspectral Proximal Sensing to Enhance Chlorophyll-a Estimation From Sentinel-2

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***Abstract*—Accurate quantification of chlorophyll-a (Chla) is vital for understanding, assessing, managing eutrophication, aquatic ecosystem health, and biogeochemical cycling. However, satellite-based Chla estimation in Case II waters remains challenging due to complex optical properties, limited synchronized field data, and uncertainties in spectral–temporal matching. This study introduces a novel approach that integrates Sentinel-2 multispectral imagery with coincident hyperspectral proximal sensing (HPS) measurements to enhance the Chla estimation accuracy in optically complex waters. First, a band-conversion fusion method with high consistency (slope = 1 and R^2 = 0.95) was proposed by leveraging strong correlations between Sentinel-2 and HPS spectra. Subsequently, a high-precision machine learning (ML)-based fused Chla model [R^2 = 0.93, normalized root-mean-squared error (NRMSE) = 24.2%, and mean absolute percentage error (MAPE) = 31.6%] was developed using a fusion dataset combining HPS measurements (N = 1728) and Sentinel-2 data (N = 100). This model significantly outperformed the unfused Sentinel-2-only model and four existing literature algorithms. When applied to Sentinel-2 time-series data (2016–2024) over Lake Taihu, the fused model revealed a notable eutrophication mitigation trend, with Chla decreasing from 30.78 to 19.56 μg/L. Comparative analyses demonstrated that fusing HPS data improved Sentinel-2-derived Chla estimation performance because strict temporal synchronization of HPS measurements reduced spectral uncertainty and enhanced sensitivity to Chla variability. Recalibrating Sentinel-2 spectra reduced MAPE from 41% to 9.7%, and expanding field-matched datasets enhanced the model reliability and robustness by 23.4% and 10.8%, respectively. The study underscores the transformative potential of multisource data fusion for advancing satellite-based water-quality monitoring, particularly in optically complex aquatic systems. The proposed framework offers a transferable methodology for estimating other biogeochemical parameters through integrated spaceborne and proximal sensing approaches, providing a foundation for improved environmental monitoring and management.**

***Index Terms*—Chlorophyll-a (Chla), data fusion, hyperspectral proximal sensing (HPS), Lake Taihu, Sentinel-2.**

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I. Introduction

PHYTOPLANKTON serves as a primary producer in aquatic ecosystems, composing the foundation of the food chain, and plays a vital role in the biogeochemical cycle of essential elements, energy flows, and ecological information transfer [1], [2]. However, human-induced eutrophication and climate change have exacerbated the frequency, duration, and intensity of phytoplankton blooms in freshwater systems globally, transforming this phenomenon into a critical challenge for aquatic ecosystem sustainability [3], [4], [5]. Especially, harmful algal blooms (HABs), characterized by the proliferation of toxin-producing species, induce cascading ecological impacts, including reduced transparency, hypoxic conditions, fish kills, and biodiversity loss [6], [7], [8]. These events directly threaten drinking water security, aquatic habitat integrity, and fisheries productivity, while posing significant risks to human health through toxin exposure via recreational activities, seafood consumption, and potable water supplies [8]. Consequently, accurate quantification of phytoplankton biomass, particularly chlorophyll-a (Chla) in aquatic ecosystems, is vital for the understanding of eutrophication, ecosystem structure and function, and the carbon cycle, as well as the reduction of health risks associated with HABs.

Chla serves as a widely accepted proxy for quantifying and tracking spatiotemporal dynamics of phytoplankton biomass in lakes [9]. Traditional field-based methods (e.g., water sampling followed by laboratory spectrophotometry or fluorometry) provide high-accuracy Chla measurements but suffer from critical limitations in large-scale or long-term monitoring: labor-intensive protocols, sparse temporal resolution, and prohibitive costs [10]. These constraints have driven the adoption of satellite remote sensing as a cost-effective, synoptic, and

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repetitive alternative for inland water-quality assessment. The spectral signature of Chla, characterized by strong absorption in blue and red wavelengths with fluorescence emission near 685 nm, enables its detection via optical remote sensing [11], [12], [13]. Since the 1980s, multispectral and hyperspectral satellites have been extensively used for Chla estimation across diverse temporal–spatial scales, such as Moderate-Resolution Imaging Spectroradiometer, Geostationary Ocean Color Imager, Landsat, Sentinel-2, and Hyperion [4], [7], [12], [14], [15], [16].

Advances in multispectral and hyperspectral sensors have driven the development of diverse Chla-retrieving algorithms leveraging spectral characteristics [17]. Current commonly quantitative Chla estimation methods for inland waters can be broadly categorized into three groups: 1) empirical algorithms derived from statistical relationships using spectral band and band combinations, 2) semi-analytical algorithms incorporating radiative transfer theory with empirical parameterization, and 3) data-driven machine learning (ML) approaches [18]. Despite these advancements, accurately and robustly estimating Chla in inland waters remains challenging. Traditional empirical algorithms, particularly those based on blue–green band ratio, perform well in phytoplankton-dominated waters but often fail in turbid environments [19], [20], where nonalgal particles and chromophoric dissolved organic matter confound spectral signals [21], [22], [23]. To address these limitations, additional empirical and semi-analytical algorithms have been proposed to enhance Chla estimation while suppressing nonalgal influences, such as red near-infrared (NIR) ratio algorithms [24], [25], [26], the normalized difference Chla index (NDCI) [23], three- and four-band algorithms [27], quasi-analytical algorithms [28], and fluorescence line height [29]. However, these methods exhibit limited transferability across different water types due to variations in model structure, spectral band selection, and inherent optical property assumptions, thereby constraining their broader applicability [30]. Recent years have witnessed a paradigm shift toward ML-based approaches, which offer superior performance in handling the nonlinear relationships and complex interactions inherent in inland water optics [15], [31]. Unlike traditional methods constrained by predefined model forms, ML algorithms adaptively learn spectral–biogeochemical patterns from data, enabling Chla estimation, such as multilayer perceptron neural network, mixture density networks, neural networks, and extreme gradient boosting (XGBoost) [21], [32], [33], [34], [35]. Among them, the XGBoost algorithm—an efficient ensemble tree model—has been widely applied to Chla monitoring due to its powerful nonlinear modeling capability, robustness to noise, high accuracy, and interpretability. For example, the XGBoost model outperformed conventional algorithms in eastern Chinese lakes [21], [35]. These advancements underscore the potential of ML to inland water-quality monitoring.

Nevertheless, high-precision satellite-based Chla estimation in turbid and eutrophic waters remains hampered by several factors. Atmospheric correction is a major bottleneck. Although numerous algorithms have been developed to counteract atmospheric interference [10], [12], spectral uncertainties introduced by inaccurate aerosol information remain relatively high in inland turbid lakes, thereby restricting the development and application of high-precision Chla estimation methods [36], [37]. Equally critical is the scarcity of synchronous Chla measurements, which undermines model stability and reliability [38], [39]. The accuracy, diversity, and representativeness of the satellite and synchronous measured datasets are the bedrock for constructing robust and high-performance water-quality estimation models [30], [40]. However, traditional manual sampling faces difficulties in acquiring sufficient synchronized data due to cloud cover and contamination. Moreover, the mismatch between the instantaneous nature of satellite imaging and the slower pace of in situ campaigns, coupled with limited financial and human resources, further constrains the number and extent of synchronized sampling [41]. To address data scarcity, many studies have expanded the temporal window between satellite images and field measurements from 3 h (h) to 7 days (d), assuming that water quality and the spectrum remain relatively stable during this period. However, it overlooks the natural variability of water bodies, which brought about a critical issue: uncertainty introduced by the temporal misalignment between the satellite-derived spectra and in situ Chla measurements. Long temporal misplacement reduces the sensitivity of spectral responses and ultimately degrades model performance [11], [41], [42]. Unfortunately, the optimal time matching window remains a contentious topic, and research on its influence on the consistency of satellite spectral and in situ data is still relatively scarce [42], [43].

In light of these challenges, integrating diverse sensors and conducting space–ground collaborative observations has become a research hotspot for enhancing the accuracy and dynamics of water environment monitoring [15], [44]. Many studies have successfully overcome the limitations of traditional satellite remote sensing in terms of time, space, and spectral resolution by synergistically utilizing the strengths of multiple data sources, providing more continuous, complete, accurate, and reliable water-quality information [45], [46], [47]. Recent advancements in hyperspectral proximal sensing (HPS)—ground-based platforms capable of acquiring continuous, high-frequency (minutely) water-leaving radiance spectra (400–1000 nm, $\Delta\lambda$ = 1 nm)—have further expanded remote sensing applications [48], [49], [50]. Unlike satellite observations, HPS spectra [HRrs(λ)] collected at a distance of 2–10 m above the water surface are unaffected by atmospheric interference and cloud cover, providing continuous, high-frequency, and accurate spectral information that is valuable for algorithm validation, mechanism understanding, and sensor calibration. However, their close-range nature limits the acquisition of large-scale spatial information. Sentinel-2, characterized by large-scale observation, short revisit period, high spatial resolution, and sensitivity bands within red-edge, has become an important tool for HABs, especially in small- and medium-sized lakes. To harness the complementary strengths of satellite- and ground-based observations, this study proposed a novel approach integrating Sentinel-2 with coincident HPS measurements to improve the Chla estimation accuracy in optically complex waters by expanding synchronized satellite and measured datasets, enhancing

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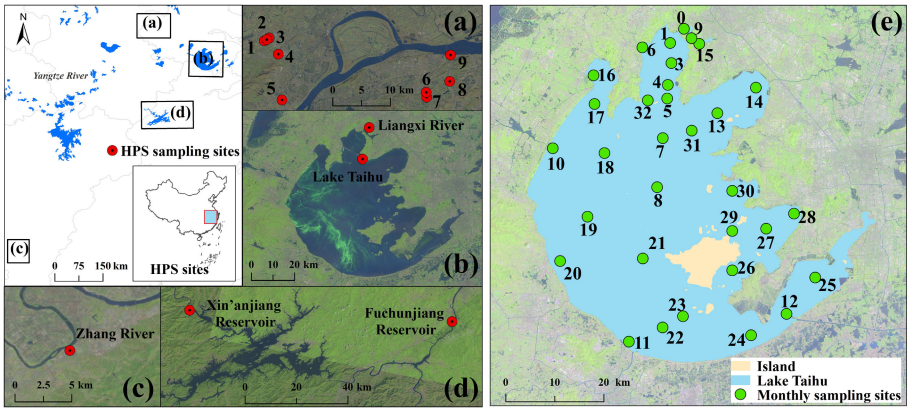


Fig. 1. Distribution of sample sites. (a)–(d) HPS sites, where the numbers 1–9 represent Baohua River, Yaochanghou River, Yaochangqian River, Zhujiashan River, Puzhen River, Lake Caiyue, Wenyuan River, Jiuxiang River, and Jiuxiang River, respectively. (e) Map of monthly monitoring sites in Lake Taihu.

spectral accuracy and sensitivity to measured Chla, and elucidating the spatiotemporal variation of Chla in Lake Taihu from 2016 to 2024. Specifically, we aimed to: 1) develop a spectral fusion method for integrating Sentinel-2 and HPS measurements; 2) construct and compare the performance of the fused Sentinel-2-HPS model, the unfused Sentinel-2 model, and four existing algorithms for Chla estimation; 3) analyze the spatiotemporal dynamics of Chla in Lake Taihu during 2016–2024; and 4) evaluate the improvements achieved by the fusion model in terms of spectral accuracy, synchronized satellite and measured matching dataset size, and the temporal matching window. This study underlined the feasibility of fusing satellite remote sensing data with HPS measurements to enhance Chla estimation in turbid waters, which offers a valuable reference for extending such approaches to the retrieval of other water-quality parameters.

II. Materials and Methods

A. Study Region

Lake Taihu (N 30.93°–31.55°, E 119.88°–120.60°), the third largest freshwater lake in China, covers an area of 2338 km^2 with an average depth of 1.9 m. It serves as a crucial drinking water source for 30 million people in the Yangtze River Delta [51], [52]. Since the 1980s, Lake Taihu has undergone rapid eutrophication and been plagued by HABs, primarily driven by anthropogenic activities and a warming climate [51], [53]. This has led to a significant transformation of the aquatic ecosystem from macrophyte-dominated to algae-dominated ecosystem, as evidenced by increasingly frequent cyanobacteria blooms [35], reduced water transparency [54], and degradation of aquatic vegetation [55], [56]. Following the severe drinking water crisis triggered by cyanobacteria blooms in 2007, the national and local governments invested approximately 100 billion Chinese yuan in a series of engineering and policy measures to control pollution and restore water quality [53]. Under these circumstances, high-precision monitoring of HABs in Lake Taihu and elucidating their temporal and spatial variations are of both scientific and practical importance. Such efforts are essential for mitigating the ecological risk and ensuring the safety of the water supply. To examine the spatial characteristics of Chla in Lake Taihu, five subregions were selected, excluding East Lake due to its extensive aquatic vegetation coverage (Fig. S1).

In addition, to enrich the diversity of water samples with different optical properties, three lakes (namely, Fuchunjiang Reservoir, Xin'anjiang Reservoir, and Caiyu Lake) and nine rivers located in the middle and lower reaches of the Yangtze River are designated as the study area (Fig. 1, Table S1).

B. Field Data

Two in situ datasets, Datasets I and II, were utilized in this study. Dataset I comprised 1906 water samples obtained from 17 field campaigns from October 2021 to November 2022. The sampling sites spanned across four lakes and nine rivers, covering a broad range of trophic states from oligotrophic to eutrophic and water depths from shallow to deep [Fig. 1(a)–(d); Table S1]. Dataset II consisted of 1980 water samples collected monthly in Lake Taihu by the Laboratory for Lake Ecosystem Research from 2016 to 2024 [Fig. 1(e)]. For both datasets, all water samples were collected using a 2.5-L acid-cleaned plastic water collector and immediately stored in the dark at 4 °C to prevent pigment degradation. Subsequently, the water samples were filtered through 0.7-μm pore-sized Whatman GF/F fiberglass filters. The pigment extraction was carried out using hot ethanol [57]. Finally, Chla concentration was determined with a Shimadzu UV2700 spectrometer at 665 and 750 nm.

C. Proximal Sensing Data

In 2018, an innovative HPS was introduced, featuring a point-type HPS instrument positioned 2–10 m above the water surface. When combined with water-quality models, it enables water-quality monitoring at a maximum frequency of 20 s [48], [58]. The HPS, equipped with a 3° field of view (FOV) and 1-nm spectral resolution spanning 400–900 nm, is deployed in the optically deep area to continuously capture water hyperspectral from 8:00 to 17:00. Compared to satellite remote sensing, its close-range observation and high-frequency monitoring allow for stronger radiance signals reception from water and more detailed depiction of short-term water-quality changes [59]. Based on the Lambertian

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whiteboard experiment, the 420–830-nm band with relative error (RE) below 3% was selected as the effective band. To minimize the impact of sun glint, sky glint, and ambient light, polarizers are used for light filtering; ambient light sensors combined with convolutional neural network algorithms eliminate skylight interference; and the ratio method with spectral vector membranes normalizes spectra without altering their shape while highlighting differences. In addition, abnormal spectra are manually removed. Further technical details of HPS are available in previous publications [49], [59], [60]. High consistency was observed between 232 pairs of HRrs(λ) and synchronous remote sensing reflectance measured by the FieldSpec spectroradiometer (Analytical Spectral Devices, Inc., Boulder, CO, USA, ASD) using the above water method under different weather conditions (sunny, cloudy, and overcast). The regression slope was 0.98, and the coefficient of determination (R^2) exceeded 0.99 ($p < 0.001$), verifying the accuracy and reliability of HPS data [49], [59], [60]. Therefore, HPS can serve as a ground-based alternative to ASD instruments for collecting hyperspectral water-reflectance measurements.

From 2020 to 2022, we collected 1736 pairs of HRrs(λ) data synchronized with in situ water samples across four lakes and nine rivers [Fig. 1(a)–(d)]. In addition, minute-level continuous HRrs(λ) data were collected in a fixed open-water site in Lake Taihu from October 2021 to September 2024 in Fig. 1(b). The measurements were conducted 4 m above the water surface, 240-m offshore at a location with a water depth of 1.75 m, where the spectral was not affected by the bottom due to the location belonging to the optically deep-water zone.

D. Sentinel-2 Data Processing

Considering sufficient spatial and temporal coverage, Sentinel-2 was selected to monitor HABs from 2016 to 2024. Sentinel-2 comprises satellite 2A (launched July 23, 2015) and satellite 2B (launched March 7, 2017), characterized by a revisit time of five days, a maximum spatial resolution of 10 m (bands 2, 3, and 8), and 13 spectral bands from visible to short-wave infrared [61]. Sentinel-2 Level-1C (L1C) and Level-2A (L2A) products from 2016 to 2024 were downloaded from the Copernicus Data Space Ecosystem (https://dataspace.copernicus.eu/). The L1C products were atmospherically corrected to L2A products using SNAP Desktop and the Sen2Cor plug-in algorithm (https://step.esa.int/main/snap-supported-plugins/sen2cor/). Subsequently, the L2A products are divided by π to obtain Sentinel-2 remote sensing reflectance [SRrs(λ)]. Cloud-contaminated pixels were removed using the cloud probability dataset provided by ESA (https://developers.google.com/earthengine/datasets/catalog/COPERNICUS_S2_CLOUD_PROBABILITY?hl = en) with a threshold of over 50. A total of 272 high-quality cloud-free images were selected based on cloud coverage of less than 2/3 (Table S2).

E. Matchup Criteria

Ideally, spectral data and in situ measurements should be collected simultaneously at the same location to develop a reliable estimation model. However, long acquisition intervals and spatial offsets often introduce inconsistencies between the obtained spectra and the actual spectra determined by water quality, particularly in lakes and rivers with complex hydrodynamic conditions and spatial heterogeneity [45]. To address these issues, the following three matchup criteria for pairing HPS data, Sentinel-2 spectral, and the measured Chla were established.

Temporally, to minimize errors arising from sampling-time discrepancies, the matching criteria for HRrs(λ) and in situ Chla measurements were set within ± 1 min (min). Spatially, water samples were collected from ± 0.1 m around the HPS sampling site to measure synchronous Chla concentration. The sampling radius (S) was calculated from the FOV of HPS and observation height using the following formula:

$$S = H \times \tan(\mathrm{FOV}/2) \tag{1}$$

where FOV is the FOV of HPS (°). S and H represent the observation radius and the height of the HPS spectrometer above the water surface (m), respectively. According to these criteria, a total of 1728 matched pairs were generated between Dataset I and HPS data, which were later fused with the matching dataset of Sentinel-2 and in situ Chla to construct the fused Chla estimation model.

Temporally, HRrs(λ) collected within ± 1 min of Sentinel-2 transit time was selected. Spatially, the Sentinel-2 spectral of the pixel corresponding to the HPS location was extracted as the matching data. After visual screening and quality control, 74 pairs of synchronous SRrs(λ) and HRrs(λ) were obtained from October 20, 2021, to December 31, 2024. This dataset was used to develop a spectral fusion method between Sentinel-2 and HPS.

For in situ Chla measurements, a temporal window of ± 2 h relative to Sentinel-2 transit was adopted to enhance data synchronization and spectral consistency. Spatially, the Sentinel-2 spectra from the pixel corresponding to each in situ sampling site were extracted for matching. By aligning Dataset II with Sentinel-2 images, a total of 118 synchronized datasets were generated. Among these, 100 pairs were employed to construct and validate the unfused Chla estimation model, while the remaining datasets were reserved for independently assessing the accuracy and applicability of both the unfused and fused Chla estimation models.

F. Spectral Simulation of Sentinel-2 Using HPS Spectral

The HPS hyperspectral data were simulated to replicate Sentinel-2 spectral bands [namely, *SHR*rs(λ)] through the spectral response functions [SRF(λ)] (https://sentiwiki.copernicus.eu/web/s2-documents) and a convolution operation, as described by the following equation:

$$SHRrs(\lambda) = \left(\int_{\lambda 1}^{\lambda 2} \mathrm{HRrs}(\lambda)\,\mathrm{SRF}(\lambda)\,d\lambda\right) \Big/ \left(\int_{\lambda 1}^{\lambda 2} \mathrm{SRF}(\lambda)\,d\lambda\right) \tag{2}$$

where HRrs(λ) denotes the original HPS hyperspectral reflectance. SRF(λ) represents the Sentinel-2 spectral response functions. $\lambda 1$ and $\lambda 2$ represent the starting and ending wavelengths for the convolution operation, respectively.

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G. Development of Chla Estimation Models

Considering the substantial regional and seasonal variability of water quality in inland waters, the optical properties exhibit pronounced spatiotemporal heterogeneity, resulting in diverse and complex relationships between spectra and water-quality parameters [62]. To address this complexity, XGBoost, a highly efficient and widely utilized ML algorithm, was employed to estimate Chla concentration due to its ability to simulate complex, nonlinear relationships and processes [21], [63]. As a gradient-enhanced tree model, it integrates multiple weak learners and iteratively constructs a robust classifier [63]. The model learns a new function to fit the residuals of the preceding tree, iteratively training newly added trees to minimize the difference between predicted and measured values until convergence. After training, k-trees are obtained, with features of training samples mapped to corresponding leaf nodes of each tree, each associated with a score. The predicted values of the training samples are derived by summing the scores from all trees. Compared with random forest models, XGBoost enhances predictive performance by progressively optimizing the model to better capture complex data relationships and feature interactions. With regularization incorporated to control model complexity and mitigate overfitting, the generalization capability of the model is improved. For hyperparameter tuning in the XGBoost model, GridSearchCV was initially employed to rapidly explore and evaluate hyperparameter combinations. Subsequently, a fivefold cross-validation method was used to evaluate and select the optimal hyperparameter set (Table S3).

To fully utilize the original spectral characteristics of inland waters, the seven single bands ranging from 443 to 783 nm were used as input features. Moreover, several commonly used spectral indices (including band ratios, band sums, band differences, and band normalizations) were introduced as candidate input features to further enhance the performance of the Chla estimation model. Correlation analysis was conducted between in situ Chla and spectral indices for both the unfused and fused training datasets (Figs. S2 and S3). Ultimately, band differences and band ratios at 705 and 665 nm, respectively, which exhibited the highest correlation coefficients (r), were selected as input features for the XGBoost algorithm.

Two synchronized datasets, the unfused (100 pairs) and fused (1828 pairs) of in situ Chla and Sentinel-2 data, were used to develop and validate Chla estimation models. The unfused synchronous dataset spanned Chla concentrations from 1.98 to 133.19 μg/L, averaging 30.35 $\pm$ 27.77 μg/L, while the fused synchronous dataset, incorporating 1728 pairs of in situ Chla and HPS data, covered a broader range (0.38–339.40 μg/L; mean $\pm$ S.D.: 46.79 $\pm$ 43.38 μg/L). Training dataset and validation datasets were randomly yielded at a 3:1 ratio from these synchronized datasets (Table S4). The unfused dataset comprised 75 samples (Chla: 1.98–133.19 μg/L; mean $\pm$ S.D.: 29.99 $\pm$ 26.52 μg/L), while the fused training dataset contained 1371 samples (0.38–320.15 μg/L; mean $\pm$ S.D.: 45.79 $\pm$ 42.82 μg/L). Correspondingly, the unfused validation datasets contained 25 samples (2.88–105.35 μg/L; mean $\pm$ S.D.: 31.42 $\pm$ 31.81 μg/L), and the fused validation dataset included 457 samples (0.76–339.40 μg/L; mean $\pm$ S.D.: 49.79 $\pm$ 44.94 μg/L). In addition, to further objectively assess the robustness and superiority of the Chla estimation model, an independent validation dataset comprising 18 water samples with diverse optical properties was collected from Lake Taihu, spanning in situ Chla concentrations of 4.52–68.92 μg/L, with an average value of 21.17 $\pm$ 21.09 μg/L.

H. Statistical Analysis

Statistical analyses, including maximum, minimum, mean, standard deviation (S.D.), and coefficient of variation (C.V.), along with Pearson correlation analysis and linear fitting, were performed using Statistical Program for Social Science (SPSS) 23.0. Results were considered statistically significant if $p \leq 0.05$. The R^2, mean absolute percentage error (MAPE), root-mean-squared error (*RMSE*), and normalized RMSE (NRMSE) were calculated to evaluate the performance and accuracy of the model. In addition, the spectral angle mapper (SAM) was introduced to evaluate spectral similarity by calculating the angular difference between the reference and the measured spectra [64], [65]. A smaller SAM value indicates greater similarity in spectral shapes. Moreover, to improve the interpretability of the XGBoost model, Shapley additive explanation (SHAP) analysis derived from game theory was used to evaluate the impact of each input feature on the model before and after data fusion. For detailed information on SHAP values, see the previous publication [66]

$$\text{C.V.} = \text{S.D.}/\text{mean} \tag{3}$$

$$\text{MAPE} = 100\% \times \frac{1}{n} \times \sum\nolimits_{i=1}^{n} (|y_i - x_i|/y_i) \tag{4}$$

$$RMSE = \sqrt{\frac{1}{n} \times \sum\nolimits_{i=1}^{n} (y_i - x_i)^2} \tag{5}$$

$$\text{NRMSE} = RMSE \Big/ \left(\frac{1}{n} \times \sum\nolimits_{i=1}^{n} y_i \right) \tag{6}$$

$$\text{SAM} = \cos^{-1} \left(\sum\nolimits_{i=1}^{n} x_i y_i \Big/ \left(\sqrt{\sum\nolimits_{i=1}^{n} x_i^2} \times \sqrt{\sum\nolimits_{i=1}^{n} y_i^2} \right) \right) \tag{7}$$

where n is the number of values; i represents the sequence number of the individual data point; and y_i and x_i represent the measured and estimated values, respectively. SAM ranges from 0° to 180°.

III. Results

A. Data Fusion of Sentinel-2 and HPS

Notable discrepancies were observed between the 74 pairs of synchronous raw Sentinel-2 spectral data and their HPS-simulated counterparts. Compared to SRrs(λ), *SHR*rs(λ) values were generally higher and more concentrated, with an average C.V. value of 28.92% $\pm$ 13.7% across bands 1–7 (Fig. 2). SAM and correlation analyses were performed for each SRrs(λ)–*SHR*rs(λ) pair. The SAM values ranged from 1.49° to 15.45°, with an average of 5.65° $\pm$ 2.72°, and over 93% were below 10°, indicating SRrs(λ) and *SHR*rs(λ) shared broadly similar spectral shapes across bands 1–7. Moreover,

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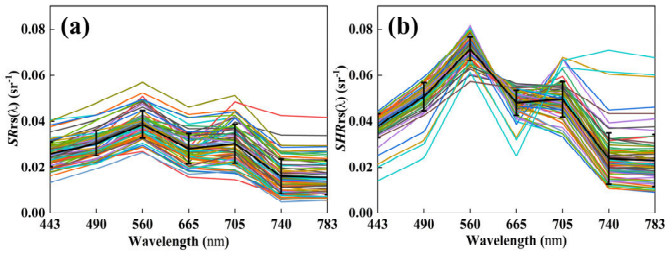


Fig. 2. Comparsion of (a) normalized SRrs(λ) and (b) synchronized *SHR*rs(λ).

the linear regression slopes for each pair ranged from 1.26 to 3.21 (averaging 2.05 $\pm$ 0.43), while the corresponding r values ranged from 0.76 to 1, averaging 0.97 $\pm$ 0.045 ($p < 0.05$), with over 94% exceeding 0.9. The high r values and wide range of slopes suggested strong linear consistency between SRrs(λ) and *SHR*rs(λ), but with notable differences in spectral amplitudes.

To eliminate observational inconsistencies and sensor-response differences between Sentinel-2 and HPS data, a band conversion method was proposed to calibrate corresponding Sentinel-2 spectra via quantitative conversion relationships, facilitating data fusion. The fusion procedure comprised two steps. First, to minimize spectral amplitude variations in SRrs(λ) caused by differing acquisition conditions, each spectral band was divided by the norm of its seven-band spectral vector to generate normalized SRrs(λ) spectra. Second, least-squares regression was applied to construct wavelength-specific conversion functions between normalized SRrs(λ) and synchronous *SHR*rs(λ) for each band, enabling further recalibration of Sentinel-2 spectra. As shown in Fig. 3, strong linear regressions were observed for all bands, with the r values ranging from 0.73 to 0.92. However, the calibration coefficients varied across bands, ranging from 0.097 to 0.16, indicating that the normalized Sentinel-2 exhibited different degrees of underestimation relative to HPS measurements at different wavelengths. Notably, the lowest R^2 of 0.54 at 443 nm (aerosol band) might be attributed to the uncertainties introduced by atmospheric conditions.

B. Improved Chla Estimation Model

XGBoost-based Chla estimation models demonstrated robust performance across both datasets, with Sentinel-2-derived Chla values and measured Chla evenly distributed along the 1:1 line [Fig. 4(a) and (b) and (d) and (e)]. Compared with the unfused Chla estimation model developed on only the Sentinel-2 dataset (R^2 = 0.91, *RMSE* = 8.77 μg/L, and NRMSE = 29.2%), the fused Chla estimation model developed on the fused dataset of HPS and Sentinel-2 performed best with the R^2 value of 0.93, *RMSE* of 11.95 μg/L, and the lowest NRMSE of 24.2% [Fig. 4(a) and (d)]. Validation results from 457 validation matchups further confirmed the superiority of the fused model, with an R^2 value of 0.90, *RMSE* of 14.82 μg/L, NRMSE of 29.8%, and MAPE of 29.8%. In contrast, the unfused Chla estimation model applied to 25 validation matchups yielded an R^2 value of 0.87, NRMSE of 36.7%, and MAPE of 63.5% [Fig. 4(b) and (e)]. For the independent validation dataset, the estimated Chla derived from the fused model demonstrated stronger agreement with measured Chla values (R^2 = 0.88, *RMSE* = 9.74 μg/L, and MAPE = 45.1%) than that derived from the unfused model (R^2 = 0.77, *RMSE* = 9.91 μg/L, and MAPE = 66.4%) in Fig. 4(c) and (f). Moreover, Fig. S4 shows a spatial comparison of the applicability and stability of the fused and unfused Chla models during HABs in Lake Taihu on June 3, 2019. The spatial distribution of the Chla map derived from the fused model Fig. S4(d) better matched the bloom features visible in the true-color Fig. S4(a) and false-color Fig. S4(b) composite images, while the unfused model failed to capture the high-Chla regions in Fig. 5(c). Higher R^2, greater spatial fidelity, and lower NRMSE and MAPE collectively indicated that the fused Chla estimation model outperformed the unfused Chla estimation model.

Then, four published models, including NIR [26], NDCI [23], three-band [27], and quasi-analytical [29], were introduced to compare the performance of the fused Chla estimation model. To ensure fair comparison and enhance model transferability, the model parameters of the NIR model, NDCI model, and three-band model were recalibrated using the same training dataset. All models exhibited improved performance on the fused dataset than on the unfused dataset, showing higher R^2 and lower NRMSE values in both the training and validation datasets in Table I. Among these, the NDCI model outperformed the four published models, with R^2 and NRMSE values of 0.85% and 36.7% on the fused training dataset, and 0.88% and 31.8% on the fused validation dataset, respectively (Table I). However, its performance was inferior to the fused XGBoost Chla estimation model, further confirming the enhanced robustness and reliability of the fused model over its unfused counterpart. Consequently, the XGBoost-based fused Chla estimation model was selected to quantify Chla dynamics in Lake Taihu, owing to its satisfactory performance, broad applicability, and strong robustness.

C. Temporal–Spatial Dynamics of Chla

Between 2016 and 2024, 272 cloud-free Sentinel-2 images over Lake Taihu were processed to generate time-series Chla products using the band conversion method and the fused XGBoost Chla estimation model. Seasonal average Chla maps were generated by averaging the same season Chla products pixel-by-pixel. Fig. 5 revealed pronounced seasonal and spatial heterogeneity in Chla concentrations across the lake, with the highest value of 140.43 μg/L recorded in the open area during summer, and the lowest value of 4.11 μg/L in Gonghu Bay during winter. Seasonal average Chla concentrations were significantly higher in summer (from June to August: 39.12 $\pm$ 13.72 μg/L) than those in spring (from March to May: 26.50 $\pm$ 8.72 μg/L) and autumn (from September to November: 30.28 $\pm$ 11.25 μg/L) ($p < 0.05$), while the lowest value was found in winter (from December to February: 14.73 $\pm$ 3.26 μg/L). Similar seasonal patterns were also recorded across the five subregions of Lake Taihu, whereas various spatial characteristics were observed. Spatially, the highest Chla concentrations in spring and summer occurred in Zhushan Bay (37.13 and 67.95 μg/L, respectively), in autumn

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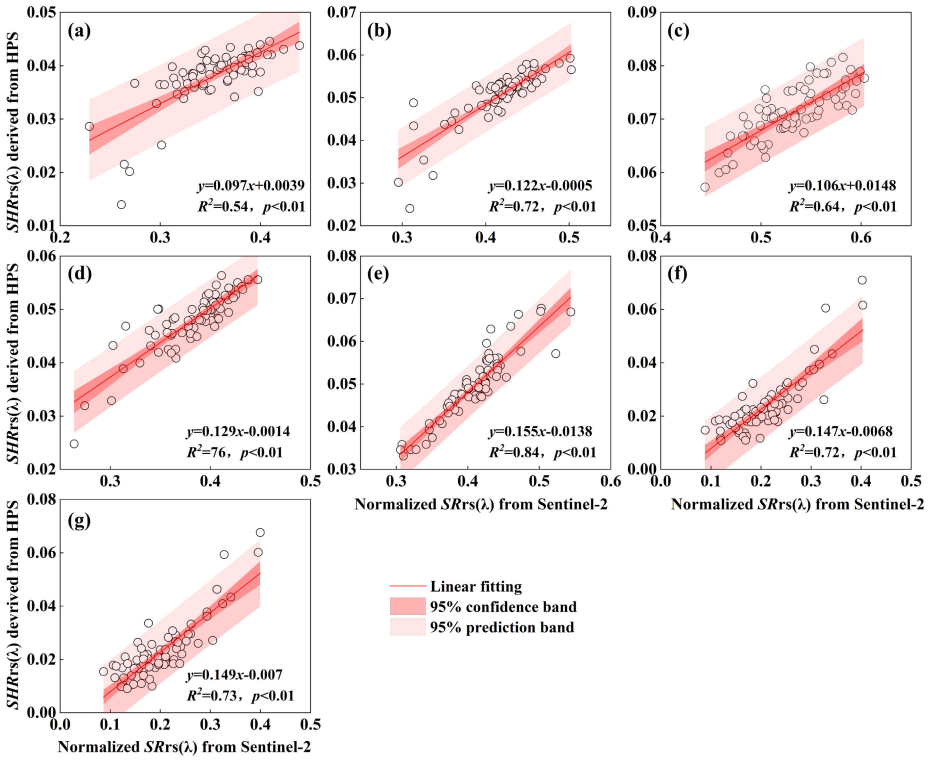


Fig. 3. Linear relationship of normalized SRrs(λ) and synchronized *SHR*rs(λ) derived from HPS at (a) 443 nm, (b) 490 nm, (c) 560 nm, (d) 665 nm, (e) 705 nm, (f) 740 nm, and (g) 783 nm.

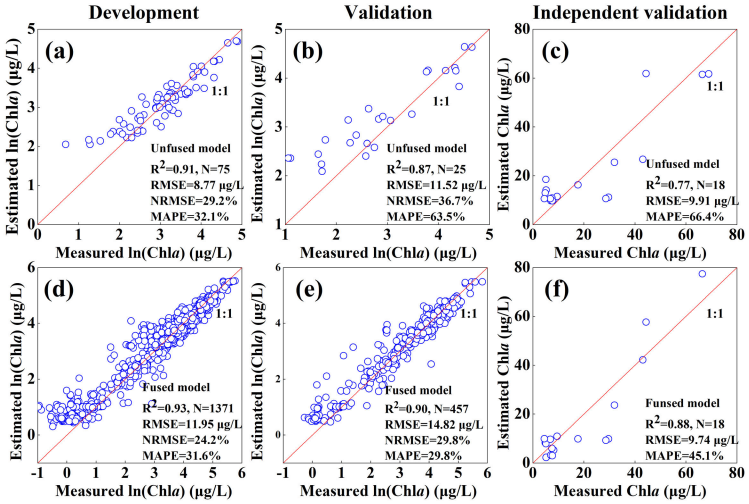


Fig. 4. (a) and (d) Development, (b) and (e) validation, and (c) and (f) independent validation of the unfused and fused Chla estimation model based on the XGBoost algorithm.

in Meiliang Bay (41.50 μg/L), and in winter in the open area (15.35 μg/L). Conversely, Xukou Bay consistently exhibited the lowest Chla concentrations across all four seasons (12.05–23.13 μg/L).

Annual average Chla maps derived from Sentinel-2 imagery were employed to analyze the interannual dynamics in Lake Taihu from 2016 to 2024. The lake-wide Chla concentration peaked at 38.29 μg/L in 2019 and reached its lowest value of 16.30 μg/L in 2023 (Fig. 6). Overall, a significant declining trend in the annual average Chla concentration was observed across the entire lake ($r = -0.76$, $p < 0.01$), ranging from 30.78 μg/L in 2016 to 19.56 μg/L in 2024, with a decrease of 36.5% and an overall average of 29.03 $\pm$ 7.79 μg/L. These results indicate effective mitigation of HABs in Lake Taihu, particularly highlighted by an annual reduction rate of 5.57 μg/L per year from 2019 to 2023 ($p < 0.01$), corresponding to a 57.4% decrease. Significant spatial and temporal variations in annual average Chla concentrations were also evident across different subregions. Most subregions peaked in 2019, except for the open area, which reached its maximum

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TABLE I
PERFORMANCE OF OUR MODEL AND PUBLISHED MODELS WITH ADJUSTED PARAMETERS BASED ON TRAINING AND VALIDATION DATASETS IN CHLA ESTIMATION

Model names	Spectral index	Training dataset				Validation dataset			
		Unfused model (N=1371)		Fused model (N=75)		Unfused model (N=457)		Fused model (N=25)	
		R^2	*NRMSE* (%)	R^2	*NRMSE* (%)	R^2	*NRMSE* (%)	R^2	*NRMSE* (%)
Our models	-	0.91	29.24	0.93	24.23	0.87	36.66	0.90	29.77
NIR ratio model	*SR*rs(705)/*SR*rs(665)	0.71	47.50	0.83	38.79	0.84	41.18	0.86	33.83
NDCI	(*SR*rs(705)-*SR*rs(665))/(*SR*rs(705)+*SR*rs(665))	0.71	46.98	0.85	36.73	0.85	40.61	0.88	31.75
Three-band model	(1/*SR*rs(665)-1/*SR*rs(705))×*SR*rs(740)	0.68	49.41	0.73	48.50	0.84	43.83	0.84	40.26
Quasi-analytical model	-	0.67	52.53	0.71	56.23	0.86	49.31	0.73	54.44

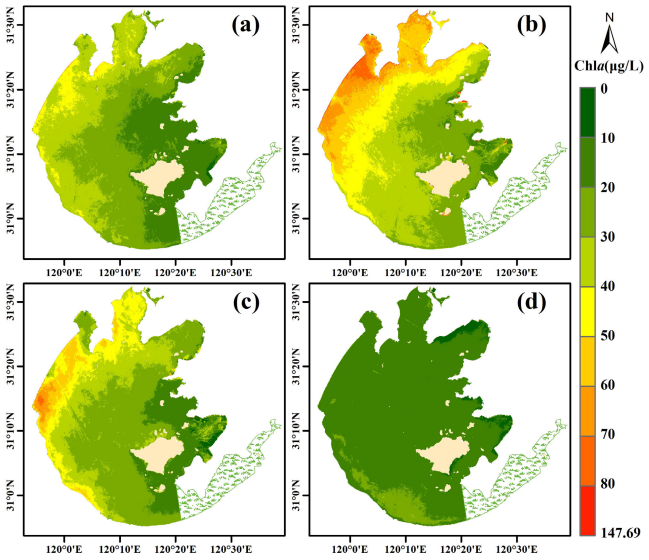


Fig. 5. (a)–(d) Seasonal average distributions of Chla in Lake Taihu from spring to winter during 2016–2024.

of 39.76 μg/L in 2017. Conversely, the lowest subregional Chla concentrations occurred in 2023 or 2024, except for Xukou Bay, where the minimum value of 14.32 μg/L was observed in 2016. These findings suggest that Lake Taihu faced severe algal bloom challenges in 2019. Over the long-term (2016–2024), significant declining trends (r ranging from −0.75 to −0.81, $p < 0.05$) in Chla were observed in four subregions of Lake Taihu, except for Xukou Bay ($r = 0.12$, p >0.05). From 2019 to 2023, Chla concentrations declined significantly across all subregions, indicating widespread alleviation of eutrophication and algal blooms. Meiliang Bay exhibited the fastest decline (8.48 μg/L per year), followed by Zhushan Bay (8.07 μg/L per year, 57.4% reduction), while Xukou Bay showed the slowest decline (1.38 μg/L per year, 29.1% reduction). Notably, Chla concentrations rebounded lake-wide in 2024, increasing by 20% relative to 2023. Similar rebounds were observed in Gonghu Bay, Xukou Bay, and the open area, suggesting that although algal bloom suppression has been effective, the risk of resurgence remains the same.

To further elucidate the spatial distribution of long-term Chla trends, a pixel-wise linear regression was conducted on annual average Chla concentrations from 2016 to 2024 (Fig. 7). Fig. 7 revealed a predominantly declining trend in Chla across Lake Taihu, with localized increases observed in Xukou Bay. From 2016 to 2024, areas with decreasing Chla trends ($r < 0$) reached 1997.44 km^2, of which 1020.93 km^2 (51.1% of the total lake area) demonstrated statistically significant declines ($r < 0$, $p < 0.05$). Significant declining trends were primarily concentrated in the open area (731.46 km^2), followed by Gonghu Bay (128.02 km^2) and Meiliang Bay (100.60 km^2). In contrast, areas with increasing Chla trends accounted for 150.55 km^2 ($r > 0$), with only 35.45 km^2 (2.3% of the increasing area) showing significant increases ($r > 0$, $p < 0.05$). Notably, 97.9% of the significantly increasing areas were located in Xukou Bay, highlighting its distinct behavior compared to other regions.

Spatially, the Sentinel-2-derived Chla data from 2016 to 2024 were averaged to generate the spatial distribution of Chla across Lake Taihu [Fig. 8(a)]. The results revealed pronounced spatial heterogeneity in Chla concentrations, with values ranging from a minimum of 6.14 μg/L in Gonghu Bay to a maximum of 118.83 μg/L in the open area, and an overall mean of 28.41 $\pm$ 8.60 μg/L. Specifically, Chla concentrations exhibited a northwest-to-southeast decreasing gradient across the lake. Zhushan Bay recorded the highest average Chla (40.77 $\pm$ 5.70 μg/L), followed by Meiliang Bay (37.92 $\pm$ 3.94 μg/L) and the open area (29.08 $\pm$ 7.57 μg/L), while Gonghu Bay (26.99 $\pm$ 5.16 μg/L) and Xukou Bay (16.92 $\pm$ 4.91 μg/L) had the lowest value. To further analyze spatial patterns of Chla variability, the C.V. was calculated using Sentinel-2-derived Chla data from 2016 to 2024 [Fig. 8(b)]. Overall, the Chla in Lake Taihu has undergone dramatic changes, with the average C.V. of 91.4%. Among them, Gonghu Bay had the strongest variation with the average C.V. of 99.6%, followed by Zhushan Bay (95.9%), while

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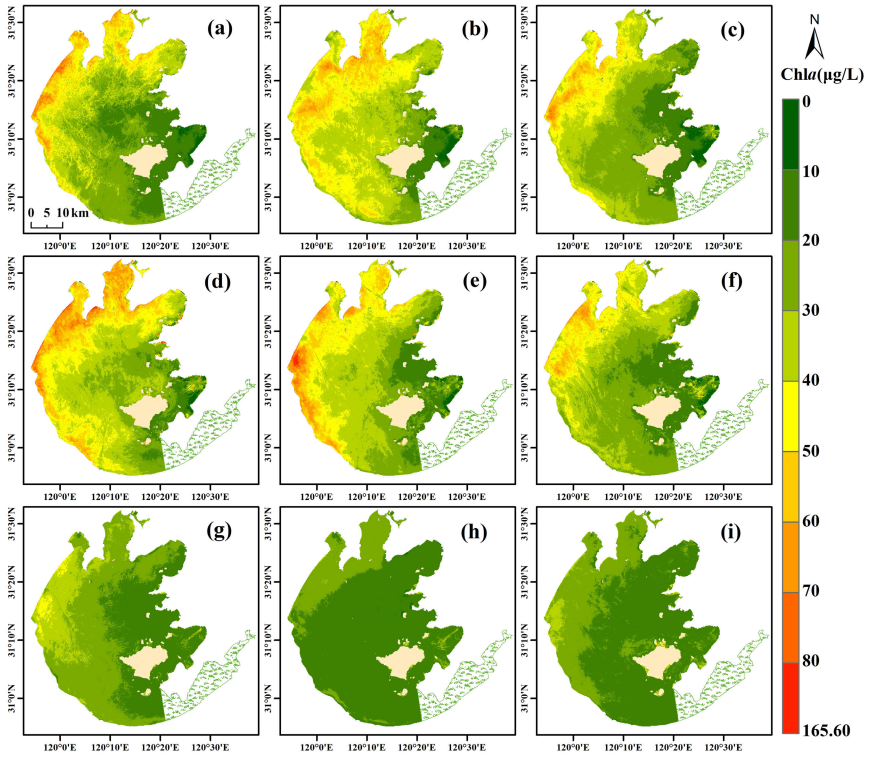


Fig. 6. (a)–(i) Maps of annual average Chla in Lake Taihu from 2016 to 2024.

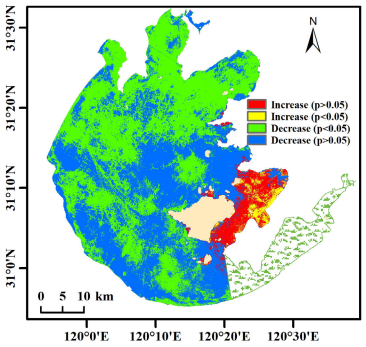


Fig. 7. Long-term trend of annual average Chla in Lake Taihu from 2016 to 2024.

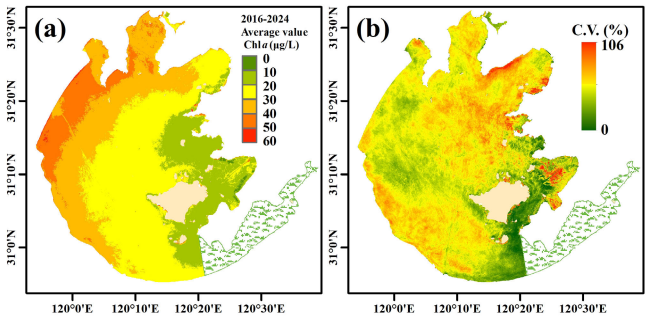


Fig. 8. (a) Spatial distribution and (b) variation of Chla in Lake Taihu from 2016 to 2024.

Xukou Bay had the smallest relative change with the average C.V. of 81%.

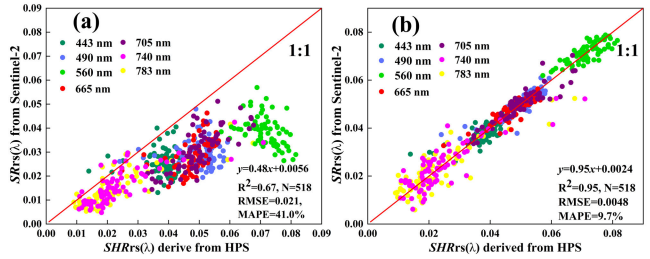


Fig. 9. Comparison of all bands of Sentinel-2 (a) before and (b) after band conversion.

IV. Discussion

A. Spectral Accuracy Assessment After HPS Data Fusion

First, Sentinel-2 spectral data were recalibrated using HPS data to minimize atmospheric correction errors, thereby improving spectral accuracy for Chla estimation. Accurate remote sensing reflectance is fundamental for high-precision water-quality remote sensing monitoring [37], [67]. However, when applied to aquatic environments, the land-oriented atmospheric correction method used in Sentinel-2 L2 products often introduces spectral bias and uncertainties, which further propagate into downstream water-quality products [37], [68], [69]. For instance, a prior study found that a 20%–30% uncertainty in reflectance of 490–743 nm can yield a 25%–70% uncertainty in Chla and total suspended matter products [37]. In contrast, in situ hyperspectral measurements acquired near the water surface are free from atmospheric interference and capture strong water-leaving signals, making them reliable

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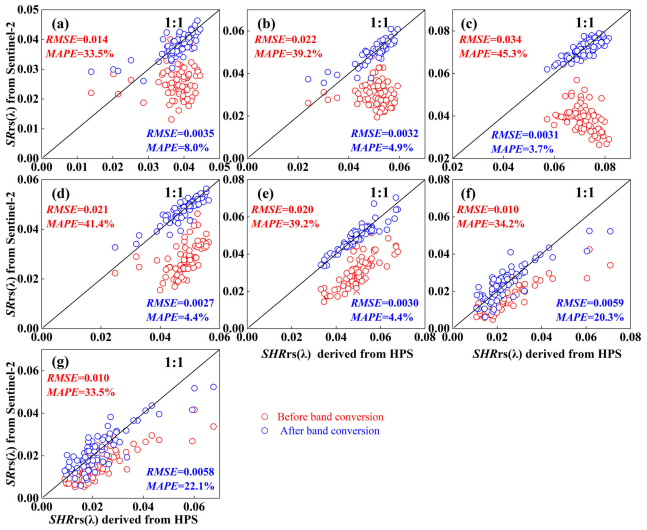


Fig. 10. Scatter plot of Sentinel-2 and HPS-simulated Sentinel-2 bands before and after band conversion at (a) 443 nm, (b) 490 nm, (c) 560 nm, (d) 665 nm, (e) 705 nm, (f) 740 nm, and (g) 783 nm.

References for evaluating atmospheric correction and spectral corrections [69], [70]. Fig. 9 illustrates a consistency analysis of 74 synchronous pairs of raw and HPS-simulated Sentinel-2 spectral bands before and after band conversion. Before conversion, SRrs(λ) was systematically underestimated relative to *SHR*rs(λ), with a linear slope of 0.48. After conversion, *RMSE* and MAPE between SRrs(λ) and *SHR*rs(λ) decreased from 0.021 to 0.0048 and from 41% to 9.7%, respectively, representing improvements of approximately 77.1% and 31.4%. The scatter point aligned closely along the 1:1 line. Moreover, SAM values ranged from 0.93° to 13.74°, with an average of 4.09° $\pm$ 2.53°, which was 27.6% lower than preconversion values, indicating remarkable improved spectral similarity between SRrs(λ) and *SHR*rs(λ) through band conversion (Fig. S5).

Further analysis revealed that the accuracy improvement between SRrs(λ) and *SHR*rs(λ) varied in different bands (Fig. 10). Except for 740 and 783 nm, with accuracy improvements of 14% and 11.3%, respectively, the other five bands improved by over 25%. Among them, the accuracy of 560 nm improved the most, reaching 41.7%, followed by 665 nm at 37%. Before band conversion, the green band (560 nm) had the highest MAPE of 45.3%, followed by the red band (665 nm) at 41.4%, while the blue band (443 nm) and the NIR band (783 nm) had the lowest MAPE values of 33.5%. After band conversion, MAPE values reached a minimum of 3.7% at 560 nm, then gradually increased to 4.4% at 665 and 705 nm, rapidly rising to 20.3% at 740 nm and 22.1% at 783 nm. The higher error in the NIR band (740 and 783 nm) likely stems from low signal strength and signal-to-noise ratio. The low spectral errors in the visible and red-edge bands provide a solid foundation for constructing high-precision Chla estimation models. In summary, reducing spectral uncertainty through band conversion significantly enhances the performance of Chla estimation models based on the fused Sentinel-2 and HPS data.

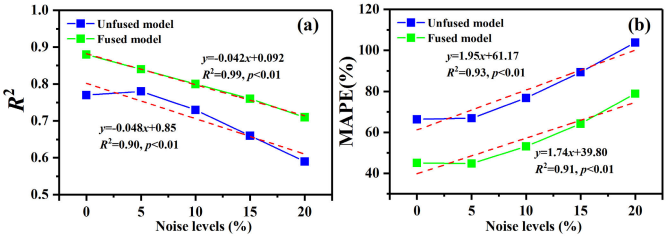


Fig. 11. (a) Variations in the coefficient of determination and (b) MAPE of the unfused model and fused model at different input spectral noise levels.

B. Reliability and Robustness of the Chla Estimation Model After HPS Data Fusion

Second, expanding the satellite–ground synchronized dataset significantly improved model reliability and robustness. The robustness and performance of reliable water-quality estimation models largely depend on the scale and quality of the satellite–ground synchronous dataset [71]. Adequate and broad synchronous datasets ensure model generalization capabilities [72]. Given the short-term variability and spectral diversity of inland water bodies, as well as spatial and temporal scales discrepancies between in situ and satellite sensors, constructing a sufficient and representative synchronization dataset for Chla remote sensing estimation is difficult but essential [32], [68]. However, for Lake Taihu, where Chla varies dramatically (Figs. 6–8), constructing a robust and stable Chla estimation model for long-term monitoring requires extensive satellite–ground synchronized training data. Unfortunately, the five-day revisit schedule of Sentinel-2, instantaneous overpass, cloud contamination, and monthly manual sampling posed challenges for acquiring sufficient satellite–ground synchronous data for Lake Taihu [48].

Therefore, to obtain sufficient satellite–ground data for ultimately constructing a robust and accurate model, we expanded the Sentinel-2 and in situ Chla synchronous datasets from 100 pairs to 1828 pairs by fusing HPS and in situ Chla datasets. Meanwhile, the Chla concentration range was also broadened from 1.98–133.19 μg/L to 0.38–339.40 μg/L. Table I showed that, regardless of linear or nonlinear models, the fused model outperformed the unfused model on the validation dataset with a higher R^2 and lower NRMSE, quantitatively demonstrating this. To further quantitatively test the stability and reliability of the fused models, Gaussian noises of 5%, 10%, 15%, and 20% were added to the input spectra of the independent validation dataset, which were then applied to the XGBoost-based fused and unfused Chla estimation models. To avoid random results, each spectrum was simulated 1000 times for each error scenario. Subsequently, R^2 and MAPE between the measured and the estimated Chla from simulated datasets were calculated. Under four noise scenarios, the accuracy of both the unfused model and fused model decreases obviously with increasing noise, with R^2 dropping from 0.77 to 0.59 and from 0.88 to 0.71, as well as MAPE increasing from 66.4% to 103.9% and from 45.1% to 78.9% (Fig. 11). Simultaneously, the unfused model consistently exhibited higher MAPE values (average difference of 23.4%) and lower R^2 values compared to the fused model, confirming the reliability of the fusion

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TABLE II
CORRELATION ANALYSIS OF SENTINEL-2 SPECTRAL AND IN SITU CHLA BEFORE AND AFTER FUSING HPS DATA

Bands (nm)	443	490	560	665	705	740	783
Non-fusion	0.013	-0.10	0.14	-0.05	-0.05	0.58	0.60
fusion	-0.73	-0.78	-0.24	-0.45	0.72	0.79	0.80

model. Linear analysis of MAPE and R^2 slopes across noise levels revealed that the unfused model (slopes: MAPE = 1.95 and R^2 = −0.048) was more sensitive to noise, while the fused model (slopes: MAPE = 1.74 and R^2 = −0.042) demonstrated greater stability against noise error propagation.

C. Effect of Time Windows on Spectral Sensitivity to Chla

Third, fusing HPS data and measured Chla within ±1 min into a satellite–ground synchronized dataset enhanced spectral sensitivity to Chla by minimizing uncertainties associated with an extended matching time window. In theory, a strict time window ensures that high spectral sensitivity to measured Chla, but an insufficient amount of synchronized data hinders robust model development [11]. Kayastha et al. [42] recommended extending the time window between Sentinel-2 and in situ Chla to five days. Although a broader time window increases the number of matchups between Sentinel-2 images and field data, it also introduces greater uncertainty due to the dynamic variability of Chla concentrations [24], [30], [68]. This is because prolonged matching windows lead to discrepancies between satellite spectra and actual Chla spectra, thereby reducing the sensitivity of satellite spectra to Chla.

To validate the aforementioned hypothesis and quantify the time window's impact on spectral consistency, we used high-frequency HPS data to set 11 matching intervals with the reference spectrum (representing Sentinel-2 spectral) taken daily at 10:40 AM. These time matching intervals ranged from ±1 to ±6 h (1-h interval), and from ±1 to ±6 d (one-day interval). SAM and MAPE values between the reference and measured Chla spectra were calculated across different time matching windows. As the time window increased, average SAM and MAPE values increased significantly from 1.79° ± 2.19° to 6.36° ± 5.07° and from 3.8% ± 5% to 13.5% ± 12.7%, respectively, indicating decreasing similarity and increasing uncertainty, weakening the correlation between Sentinel-2 spectra and measured Chla (Fig. 12). Within the first 6 h, MAPE rapidly increased to 9.8%, averaging an increase of 1.1%/h, while it stabilized at around 13% for time windows >3 days [Fig. 12(b)]. Commonly adopted time matching windows of 2 h, 3 h, and 1 d resulted in spectral error of 5.6%, 6.5%, and 10.2%, respectively. In addition, correlation coefficients of corresponding bands under different time windows were calculated and compared (Fig. 13). Although correlation coefficients of all bands decreased significantly with a wider time window, the responses varied across different bands. The decrease was most pronounced at 560 nm (slope = −0.077), followed by 665 and 740 nm (slope = −0.072), while the smallest decrease was observed at 705 nm (slope = −0.047), which were sensitive to the absorption and scattering of Chla (Fig. 13). Finally, comparing Sentinel-2 and measured Chla data before and after HPS fusion showed that the post-fusion correlation coefficient was significantly higher than that the prefusion value (Table II).

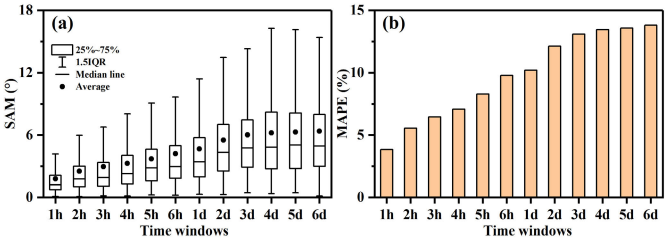


Fig. 12. Variations in (a) SAM and (b) MAPE under different matching time windows.

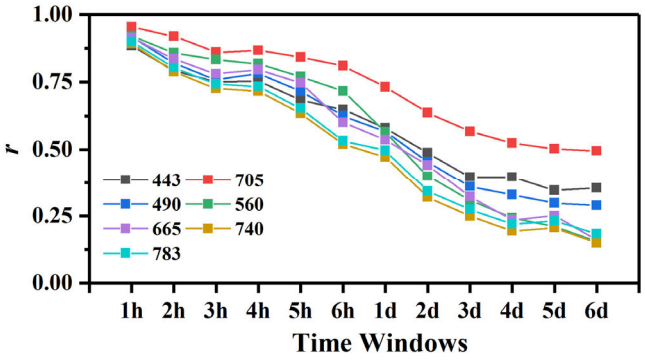


Fig. 13. Correlation coefficient between $SHRrs(\lambda)$ and in situ Chla under different matching time windows.

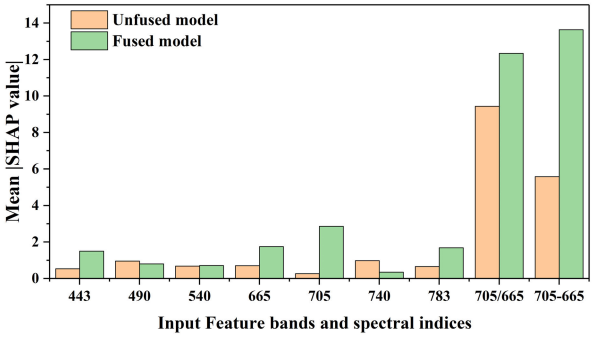


Fig. 14. Average impact of input features on unfused and fused Chla estimation model.

D. Comprehensive Impact on Model Performance After HPS Data Fusion

To comprehensively quantify the interpretability of data fusion on model improvement, we introduced SHAP analysis to evaluate the impact of each input feature on the model. As shown in Fig. 14, regardless of the unfused and fused Chla estimation models, the difference and ratio between SRrs(705) and SRrs(655) are the input features that contribute most. Compared with the unfused model, except for SRrs(490) and SRrs(740), the mean absolute SHAP values of most input features of the fused model have significantly increased, indicating that the fusion of HPS data enhanced their explanatory power and the model's dependence on these

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input features. Among them, the difference and ratio between SRrs(705) and SRrs(655) increased the most and the second most (8.04 and 2.90, respectively), followed by SRrs(705) (2.59), revealing that the fusion process enhances the potential relationships between these sensitive features and Chla concentration. Conversely, the reduced importance of features such as 490 and 740 implies that their contributions were partially replaced by more informative features. Therefore, integrating HPS data further highlights the impact of key input features sensitive to Chla, thereby improving the overall accuracy and interpretability of the model.

E. Uncertainty of Data-Fused Chla Estimation Framework

Although the enhanced performance of the Chla estimation model achieved by fusing HPS data and Sentinel-2 data, several sources of uncertainty persist. First, the discrepancy in spatial resolution between the two datasets was overlooked during fusion. HPS data, collected by a single-point sensor, represent the water spectra near the sampling point [59], while Sentinel-2 data represented pixel spectra with a resolution of 10×10 m. Considering the heterogeneity of water bodies and the presence of mixed pixels, aligning HRrs(λ) with SRrs(λ) may result in mismatches, thereby introducing errors. This error is currently unavoidable and represents a common challenge in satellite remote sensing. Second, the regional and limited synchronous HPS and Sentinel-2 datasets may introduce errors in the spectral conversion process. It should be noted that the current band conversion method (Fig. 3) and relevant results (Figs. 9 and 10) were derived from matchups ($N = 74$) between Sentinel-2 and HRrs(λ) in Lake Taihu. Consequently, it may limit its application in other regions. To address this, deploying more HPS equipment to collect additional spectra synchronized with Sentinel-2 will be crucial for developing a more robust spectral fusion method in the future.

F. Implications for Water Environmental Management

The increasing global frequency and magnitude of HABs [4], [7] highlight the importance of long-term, timely, and high-precision monitoring of Chla dynamic monitoring for eutrophication control and aquatic ecosystem restoration [73]. However, accurately monitoring Chla in inland waters remains challenging due to their complex optical properties. Satellite remote sensing offers a valuable solution by providing large-scale, cost-free, and long-term Chla monitoring data, supporting research on bloom dynamics and their responses to human activities and climate change [38], [74]. Nevertheless, limitations in atmospheric correction accuracy, insufficient satellite and in situ synchronous data, and uncertainties arising from temporal matching windows between satellite and in situ data constrain the accuracy of Chla estimation. These constraints may further impact the implementation of scientific policies [11], [75]. This study employed data assimilation techniques to fuse HPS data with Sentinel-2 data, thereby improving satellite spectral accuracy, expanding the satellite and in situ synchronous dataset, reducing uncertainty caused by time windows, enhancing spectral sensitivity to Chla, and ultimately improving the accuracy and robustness of the Chla estimation model. This approach revealed a reduction in algal blooms in Lake Taihu from 2016 to 2024.

Our findings have two key implications for water environmental management. First, this study provides a long-term, high-precision Chla dataset for assessing water eutrophication and understanding the response of algal blooms to climate change and human activities. This, in turn, assists in formulating more effective and scientifically grounded water environmental management strategies. In the future, the fused Chla model will be applied to large-scale Sentinel-2 and high-frequency HPS datasets to establish a 3-D monitoring network. This network will provide comprehensive data support for water environment monitoring, assessment, restoration, and management from local to lake-wide scales and from static to dynamic perspectives.

Second, the data fusion approaches employed in this study can be extended to other satellites and ground-based hyperspectral sensors to enhance the monitoring accuracy of other water-quality parameters, such as total suspended matter and water transparency. Integrating the strengths of different sensors and conducting space–ground collaborative observations are current research hotspots to enhance the accuracy and dynamism of environment monitoring [15], [44], [75], [76], [77]. With the global launch of numerous satellites for water body and ocean monitoring, and the increasing application of ground-based hyperspectral sensors in water body monitoring driven by technological advancements, exploring more data assimilation and fusion methods will be crucial for enhancing the accuracy of satellite-based water-quality monitoring in the future.

V. Conclusion

In this study, we proposed a novel approach by fusing HPS data with Sentinel-2 multispectral imagery to enhance Chla estimation accuracy, enabling refined spatiotemporal variation of Chla in Lake Taihu from 2016 to 2024. For data assimilation and fusion, we employed a band conversion method, leveraging the strong linear consistency between synchronous HPS and Sentinel-2 data. This spectral alignment formed the basis for training and validating an ML-based Chla estimation model using paired datasets: Sentinel-2 imagery ($N = 100$) and in situ Chla measurements ($N = 1728$), both before and after HPS data fusion. The fused model demonstrated superior performance across a broader Chla concentration range (0–200 μg/L), outperforming both the Sentinel-2-only model (MAPE reduced by 36.7%) and four established algorithms from the literature. Time-series application of this model to Sentinel-2 data (2016–2024) revealed a significant decline in Chla concentrations (from 30.78 to 19.56 μg/L) in Lake Taihu, indicating marked alleviation of cyanobacterial bloom intensity. Comprehensive analyses validated three key improvements from HPS and Sentinel-2 fusion: 1) enhanced spectral accuracy through atmospheric correction bias calibration (MAPE reduced from 41% to 9.7%); 2) improved model stability via expanded satellite–ground synchronous datasets; and 3) heightened spectral sensitivity to Chla variability through strict temporal synchronization. These synergistic

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effects collectively reduced model uncertainty and enhanced Chla estimation accuracy. This study underscores the feasibility and advantages of fusing satellite multispectral data with HPS measurements for accurate Chla estimation in turbid Case II waters. The proposed framework offers transferable methodologies for enhancing satellite-based monitoring of water-quality parameters through multisource data integration, providing critical insights for environmental management and ecosystem health assessment.

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